Unit I Analytical Assignments with SDGs

Course: ITA0614 - Machine Learning

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1. Explain how Machine Learning can identify poverty-prone regions using socio-economic data.

Detailed Explanation: Machine Learning models, specifically classification algorithms, can identify poverty-prone zones by processing multifaceted socio-economic indicators such as household income, literacy rates, and local unemployment metrics. By recognizing non-linear relationships within these historical datasets, the model mathematically segregates regions, allowing policymakers to deploy targeted interventions to vulnerable areas efficiently.

Algorithm Identification: **Decision Tree (ID3)**

Expanded Numerical Calculation:

1. Dataset Definition: We evaluate 10 regional samples, consisting of 6 recognized Poverty zones and 4 Non-Poverty zones.

2. Initialization: The initial entropy of the dataset prior to any attribute split is mathematically calculated to be 0.971.

3. Step-by-Step Execution: We evaluate a split based on the socio-economic feature 'Low Income (Yes/No)'.
- Left Branch (Low Income = Yes): Groups 6 regions (5 Poverty, 1 Non-Poverty). The entropy of this branch is 0.650.
- Right Branch (Low Income = No): Groups 4 regions (1 Poverty, 3 Non-Poverty). The entropy of this branch is 0.811.

4. Information Gain Calculation:
- Formula: Parent Entropy - Weighted Child Entropy
- 0.971 - [(6/10 × 0.650) + (4/10 × 0.811)]
- 0.971 - [0.390 + 0.324] = 0.971 - 0.714 = 0.257

5. Final Conclusion: The 'Low Income' feature yields a positive Information Gain of 0.257, proving it is a mathematically significant attribute for accurately categorizing a region as poverty-prone.

2. Explain how Machine Learning can predict crop yield using historical agricultural data.

Detailed Explanation: Predictive agricultural models utilize historical environmental metrics—such as annual rainfall, average temperature, and soil pH levels—as input variables to forecast continuous output values representing crop yield. By establishing a mathematical mapping function, these models enable farmers to make data-driven decisions regarding resource allocation and harvesting schedules based on current climatic conditions.

Algorithm Identification: **Simple Linear Regression**

Expanded Numerical Calculation:

1. Dataset Definition: We analyze historical records over three distinct seasons. Rainfall X (mm) = [10, 20, 30]. Corresponding Yield Y (tons) = [50, 70, 90].

2. Mathematical Formula: We aim to fit the data to the linear equation Y = mX + c, where 'm' represents the slope (rate of change) and 'c' represents the Y-intercept.

3. Step-by-Step Execution:

- Calculate the slope (m) using two data points: m = (Y2 - Y1) / (X2 - X1).
- m = (70 - 50) / (20 - 10) = 20 / 10 = 2.
- Calculate the intercept (c) by substituting a point into the equation: 50 = 2(10) + c.
- Therefore, solving for c yields: c = 30.
- The final trained predictive model equation is: Y = 2X + 30.

4. Final Conclusion: For an upcoming agricultural season with a forecasted 40mm of rainfall, the model mathematically predicts a successful crop yield of Y = 2(40) + 30 = 110 tons.

3. Explain the role of Machine Learning in disease prediction and early diagnosis.

Detailed Explanation: In the healthcare sector, Machine Learning facilitates early disease prediction by analyzing historical patient records and finding consistent diagnostic patterns. By mapping clinical symptoms (like fever or blood pressure anomalies) to specific medical conditions, these concept learning algorithms automate the diagnostic process, drastically reducing human error and expediting critical care pathways for at-risk patients.

Algorithm Identification: **Find-S (Concept Learning)**

Expanded Numerical Calculation:

1. Dataset Definition: Two patient records are evaluated for a specific positive disease marker. Patient 1 exhibits (Fever = High, Cough = Yes). Patient 2 exhibits (Fever = High, Cough = No).

2. Initialization: The hypothesis boundary is strictly initialized to the most specific possible state: S0 = ⟨∅, ∅⟩.

3. Step-by-Step Execution:

- Process Patient 1 (Positive): The specific boundary generalizes to exactly match this first instance, becoming S1 = ⟨High, Yes⟩.
- Process Patient 2 (Positive): The algorithm compares S1 to the new instance. The first attribute ("High") matches perfectly. The second attribute ("Yes" vs "No") conflicts.
- To maintain consistency covering both positive instances, the conflicting attribute must generalize to the wildcard symbol '?'.

4. Final Conclusion: The updated specific hypothesis is S2 = ⟨High, ?⟩, demonstrating mathematically that high fever is the primary disease marker, completely independent of the presence of a cough.

4. Explain how recommendation systems improve online learning platforms.

Detailed Explanation: Recommender systems leverage student behavioral data—such as course completion histories, viewing durations, and topic preferences—to deliver highly personalized educational content. By mathematically measuring the similarity between users, collaborative filtering algorithms can accurately predict and suggest new courses that align with a student's specific learning trajectory, thereby significantly boosting platform engagement.

Algorithm Identification: **K-Nearest Neighbors (KNN) Collaborative Filtering**

Expanded Numerical Calculation:

1. Dataset Definition: The objective is to recommend a new course to Student A. Student A's history includes [Python, Machine Learning]. Student B's history includes [Python, Machine Learning, Deep Learning]. Student C's history includes [Java, C++].

2. Initialization/Formula: We utilize the Jaccard Similarity index (Intersection over Union) to measure mathematical proximity between the students.

3. Step-by-Step Execution:

- Calculate Similarity between A and B: Intersection = 2 (Python, ML). Union = 3 (Python, ML, Deep Learning). Sim(A,B) = 2/3 = 0.667.
- Calculate Similarity between A and C: Intersection = 0. Union = 4. Sim(A,C) = 0/4 = 0.000.
- Identify Nearest Neighbor: The maximum similarity score definitively belongs to Student B (0.667).

4. Final Conclusion: Since Student B is the nearest mathematical neighbor to Student A, and Student B has completed "Deep Learning", the platform will automatically recommend the "Deep Learning" course to Student A.

5. Develop a Machine Learning model to detect water contamination.

Detailed Explanation: Environmental Machine Learning models function as automated anomaly detection systems for water quality management. By ingesting continuous chemical sensor data—including pH levels, turbidity, and heavy metal concentrations—a probabilistic classifier can evaluate the readings against historical safety thresholds, instantly flagging samples that pose a biological or chemical hazard to public health.

Algorithm Identification: **Naive Bayes Classifier**

Expanded Numerical Calculation:

1. Dataset Definition: A historical database of 10 water samples reveals 3 were 'Contaminated' and 7 were 'Safe'. A specific chemical feature, High Lead Level (>15 ppb), was present in 2 of the Contaminated samples and 1 of the Safe samples.

2. Initialization/Formula: Establish Prior Probabilities: P(Contaminated) = 3/10 = 0.300. P(Safe) = 7/10 = 0.700.

3. Step-by-Step Execution:
- Calculate Likelihoods for the 'High Lead' feature: P(High Lead | Contaminated) = 2/3 ≈ 0.667. P(High Lead | Safe) = 1/7 ≈ 0.143.
- Evaluate a new incoming sample that registers 'High Lead'.
- Score for Contaminated = P(Contaminated) × P(High Lead | Contaminated) = 0.300 × 0.667 = 0.200.
- Score for Safe = P(Safe) × P(High Lead | Safe) = 0.700 × 0.143 = 0.100.

4. Final Conclusion: Comparing the probabilistic scores, 0.200 is strictly greater than 0.100. The algorithm conclusively classifies the new water sample as Contaminated, triggering an automatic safety alert.